

Modes of DES (Data Encryption Standard)

Aim

To implement the various **Modes of DES (Data Encryption Standard)** namely **ECB, CBC, CFB, OFB, and CTR** for secure data encryption and decryption, and to compare their working principles.

Algorithm

Step 1:Start the program.

Step 2:Read the plaintext from the user.

Step 3:Read the secret key.

Step 4:Display the available DES modes:

- ECB
- CBC
- CFB
- OFB
- CTR

Step 5:Select the desired mode.

Step 6:Perform encryption according to the selected mode.

- **ECB:** Encrypt each block independently.
- **CBC:** XOR each plaintext block with the previous ciphertext block before encryption.
- **CFB:** Encrypt the previous ciphertext block and XOR it with the plaintext.
- **OFB:** Encrypt the previous output block and XOR it with the plaintext.
- **CTR:** Encrypt the counter value and XOR it with the plaintext.

Step 7:Display the encrypted ciphertext.

Step 8:Perform decryption using the selected mode.

Step 9:Display the recovered plaintext.

Step 10: Stop the program.

C Program (Modes of DES Demonstration)

```
#include <stdio.h>
#include <string.h>

void ECB(char text[], char key[])
{
    int i;
    printf("\nECB Cipher: ");
    for(i=0;text[i]!='\0';i++)
```

```

        printf("%02X", text[i]^key[i%strlen(key)]);}

void CBC(char text[], char key[]){
    int i, iv=10;

    char c;

    printf("\nCBC Cipher: ");
    for(i=0;text[i]!='\0';i++){
        c=(text[i]^iv)^key[i%strlen(key)];
        printf("%02X", c);
        iv=c;} }

void CFB(char text[], char key[]){
    int i, iv=20;

    char c;

    printf("\nCFB Cipher: ");
    for(i=0;text[i]!='\0';i++) {
        c=text[i]^(iv^key[i%strlen(key)]);
        printf("%02X", c);
        iv=c;} }

void OFB(char text[], char key[]){
    int i, iv=30;

    char out;

    printf("\nOFB Cipher: ");
    for(i=0;text[i]!='\0';i++){
        out=iv^key[i%strlen(key)];
        printf("%02X", text[i]^out);
        iv=out;} }

void CTR(char text[], char key[]){
    int i, counter=1;

    char out;

    printf("\nCTR Cipher: ");
    for(i=0;text[i]!='\0';i++)
    {
        out=(counter++)^key[i%strlen(key)];

```

```

        printf("%02X", text[i]^out);} }

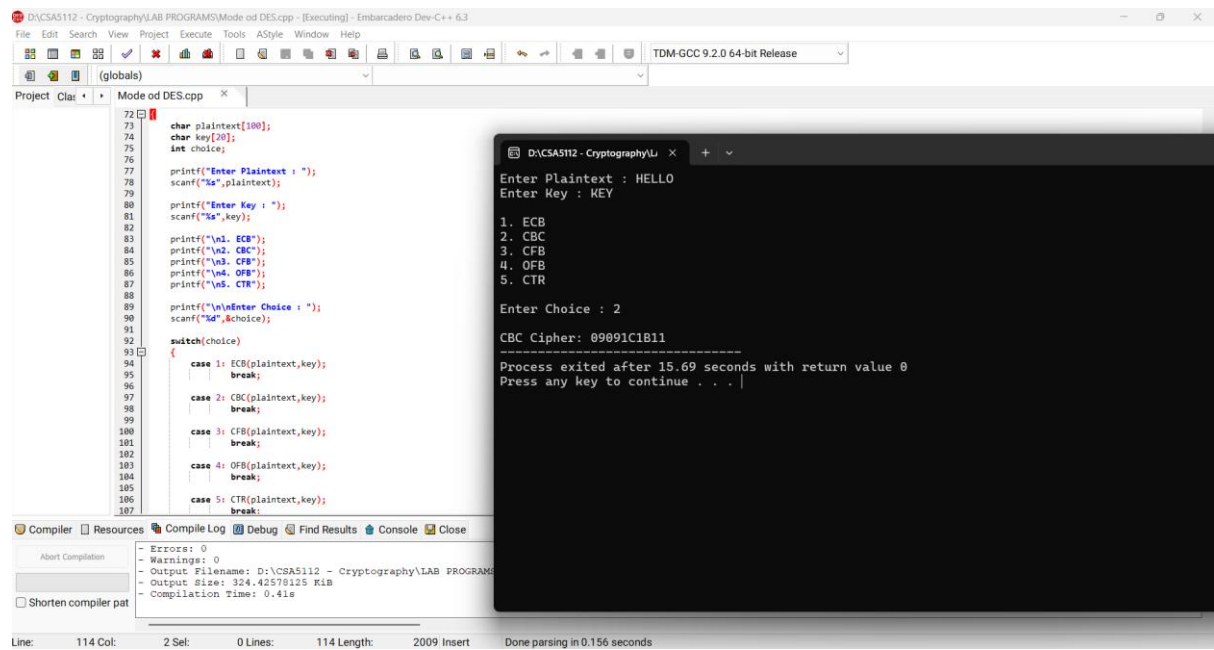
int main(){
    char plaintext[100];
    char key[20];
    int choice;
    printf("Enter Plaintext : ");
    scanf("%s",plaintext);
    printf("Enter Key : ");
    scanf("%s",key);
    printf("\n1. ECB");
    printf("\n2. CBC");
    printf("\n3. CFB");
    printf("\n4. OFB");
    printf("\n5. CTR");
    printf("\n\nEnter Choice : ");
    scanf("%d",&choice);

    switch(choice)
    {
        case 1: ECB(plaintext,key);
            break;
        case 2: CBC(plaintext,key);
            break;
        case 3: CFB(plaintext,key);
            break;
        case 4: OFB(plaintext,key);
            break;
        case 5: CTR(plaintext,key);
            break;
        default:
            printf("Invalid Choice"); }

    return 0;}

```

OUTPUT :



The screenshot displays an IDE window titled "D:\CSA5112 - Cryptography\LAB PROGRAMS\Mode od DES.cpp - [Executing] - Embarcadero Dev-C++ 6.3". The code in the editor is as follows:

```
72 char plaintext[100];
73 char key[20];
74 int choice;
75
76
77 printf("Enter Plaintext : ");
78 scanf("%s",plaintext);
79
80 printf("Enter Key : ");
81 scanf("%s",key);
82
83 printf("\n1. ECB");
84 printf("\n2. CBC");
85 printf("\n3. CFB");
86 printf("\n4. OFB");
87 printf("\n5. CTR");
88
89 printf("\nEnter Choice : ");
90 scanf("%d",&choice);
91
92 switch(choice)
93 {
94     case 1: ECB(plaintext,key);
95             break;
96     case 2: CBC(plaintext,key);
97             break;
98     case 3: CFB(plaintext,key);
99             break;
100    case 4: OFB(plaintext,key);
101            break;
102    case 5: CTR(plaintext,key);
103            break;
104    }
105
106
107
```

The output window shows the following execution results:

```
D:\CSA5112 - Cryptography\LAB PROGRAMS\Mode od DES.cpp - [Executing]
Enter Plaintext : HELLO
Enter Key : KEY

1. ECB
2. CBC
3. CFB
4. OFB
5. CTR

Enter Choice : 2

CBC Cipher: 00091C1B11
-----
Process exited after 15.69 seconds with return value 0
Press any key to continue . . .
```

The IDE's status bar at the bottom indicates "Line: 114 Col: 2 Sel: 0 Lines: 114 Length: 2009 Insert Done parsing in 0.156 seconds".